

ARTIFICIAL INTELLIGENCE

Lecture 02: Introduction to AI, Human Perception & Logic

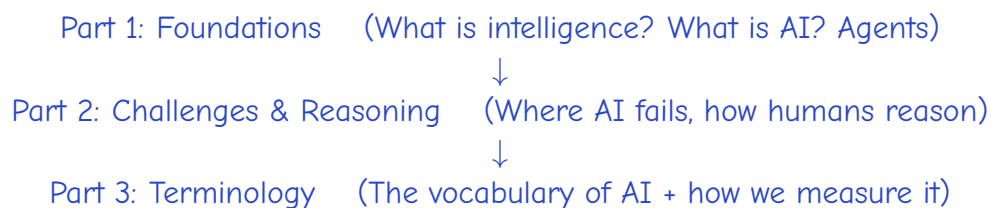
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(CIS313 — Daffodil International University, Fall 2026)

What this lecture teaches:

1. What intelligence is, and how human intelligence differs from artificial intelligence.
2. The 5 components (building blocks) of AI.
3. Agents and environments — the PEAS description and types of environments.
4. The challenges and limitations of today's AI.
5. How humans turn raw perception into logical sentences — and why AI finds it hard.
6. Core and advanced AI terms: neural networks, ML, deep learning, perceptron, fuzzy logic ...
7. Evaluation metrics (accuracy, precision, recall, F1), gradient descent, and the Turing Test.

Big Picture of this Lecture



Basic Words You Need First (Zero Knowledge Needed)

Word	Simple meaning
Data	Facts or information (numbers, words, pictures, sounds) that a computer can store.
Dataset	A big collection of data used to teach an AI (e.g. 1 million photos of cats and dogs).
Label	The correct answer attached to a piece of data (photo → “cat”). Labeled data has answers; unlabeled data doesn’t.
Model	The “brain” of an AI — the thing that has learned from data and now makes predictions.
Training	The process of teaching a model by showing it lots of data.
Prediction	The model’s answer for new data (“this email is spam”).
Pattern	Something that repeats in data (e.g. spam emails often contain “FREE!!!”).
Sensor	A device that collects information from the world (camera, microphone, thermometer).
Algorithm	A step-by-step recipe to solve a problem.
Bias	An unfair tendency — the AI treats some groups or cases unfairly because of what it learned.
Error / Loss	How wrong the model is. Smaller = better.

1. What is Intelligence?

Intelligence = the ability to **acquire** (get), **process** (think about), and **apply** (use) knowledge to **solve problems** and **adapt to new situations**.

Break it down:

- **Acquire knowledge** → learn something new (read, see, hear, experience).
- **Process knowledge** → understand it, connect it to what you already know.
- **Apply knowledge** → use it to do something useful.
- **Adapt** → handle a situation you have **never seen before**.

Think of it like this (real life):

You learn to ride a bicycle (**acquire**). You understand that leaning left turns you left (**process**). You ride to university (**apply**). One day the road is wet and slippery — you ride slower and brake earlier, even though nobody taught you “wet road rules” (**adapt**). That is intelligence.

Human Intelligence vs Artificial Intelligence

Trait	Human Intelligence	Artificial Intelligence
Learning	Learns from a few examples and lived experience	Needs large datasets (labeled or unlabeled)
Emotion & context	Reads sarcasm, tone, body language	Struggles without clear signals
Speed & scale	Limited by biology; gets tired	Millions of calculations per second; never tired
Creativity	Original ideas from lived experience	Recombines patterns seen in training data
Consistency	Changes with mood, bias, tiredness	Same input → same output, every time

Each trait explained

1. Learning.

A child sees **2–3 cats** and can recognise any cat for life. An AI may need **thousands** of cat photos to do the same.

2. Emotion & context.

Humans understand what is *meant*, not only what is *said*. AI mostly sees only the words or pixels.

Example:

A friend rolls their eyes and says: “Great, *ANOTHER* meeting.”

- A **human** instantly understands the **sarcasm** → the friend is annoyed.
- A basic **AI text classifier** (a program that labels text as positive/negative), trained only on words, sees the word “Great” and may label it **positive**. It cannot see the eye-roll or hear the tone.

3. Speed & scale.

A human can check maybe 50 exam papers in a day and gets tired. A computer can check millions of records per second, all night.

4. Creativity.

Humans create from experience and feeling. AI creates by **mixing patterns** it has already seen — it can produce something new-looking, but it is built from its training data.

5. Consistency.

A teacher might mark more strictly when tired. An AI gives the **same answer** to the same input every time (good for fairness, but it also repeats the same mistake every time).

Simple:

Humans are better at understanding, context, and learning from few examples. AI is better at speed, scale, and consistency.

2. What is Artificial Intelligence?

Artificial Intelligence (AI) = the **simulation** of human intelligence in machines, designed to perform tasks that normally need human thinking — **learning, reasoning, problem-solving, and decision-making**.

Simulation = copying or imitating. AI *imitates* intelligent behaviour; it does not necessarily think like a human.

AI Around Us — 6 Real Applications

Application	How AI is used
Voice Assistants	Siri, Google Assistant, Alexa use speech recognition (sound → words) and NLP (understanding the words) to answer spoken questions.
Recommendations	Netflix and Spotify suggest shows/songs using your viewing history and the patterns of similar users (“people like you also watched...”).
Autonomous Vehicles	Tesla Autopilot and Waymo combine cameras, radar and lidar to steer, brake, and change lanes by themselves.
Robotics	Boston Dynamics’ Spot and Amazon’s warehouse robots move around and pick up objects in the real world.
NLP & Chatbots	ChatGPT-style assistants write text, translate languages, and hold conversations.
Medical Diagnostics	AI models spot tumours in X-ray / MRI scans and support doctors (e.g. IBM Watson Health).

Lidar = a sensor that shoots laser light and measures how long it takes to bounce back, to build a 3D map of surroundings.

Simple:

AI = machines copying human thinking skills to do useful jobs.

3. The Five Components of AI

AI is built from **5 building blocks**. Most real AI systems use several of them together.

Think of it like this (real life):

Think of a **human doctor**. The doctor **knows** medical facts (knowledge), has **learned** from years of patients (learning), **reasons** “fever + cough → maybe flu” (reasoning), **looks and listens** to the patient (perception), and **talks** to the patient (language). AI tries to build each of these 5 skills into a machine.

1. Knowledge Representation

Knowledge representation = structuring facts about the world in a form a computer can store and reason with.

Example:

A **medical ontology** links “fever” → its possible causes (infection, flu, malaria ...).
(**Ontology** = an organised map of concepts and how they are related.)

Simple:

Storing facts in an organised way so the computer can use them.

2. Machine Learning (ML)

Machine learning = learning patterns from data, instead of being given fixed rules by a programmer.

Example:

A **spam filter** learns what spam looks like from **millions of labeled emails** (“spam” / “not spam”). Nobody writes every rule; the filter discovers the patterns itself.

Simple:

Learning from examples.

3. Reasoning & Problem Solving

Reasoning = drawing logical conclusions from known facts, and searching for a solution to a problem.

Example:

A **chess engine** searches through possible moves to find a checkmate.
(This is exactly the **search** from Lecture 0!)

Simple:

Thinking step by step to reach an answer.

4. Perception

Perception = interpreting input from sensors — seeing, hearing, touching.

Example:

Face-unlock on your phone recognises you from a camera image.

Simple:

Making sense of what the machine sees or hears.

5. Natural Language Processing (NLP)

NLP = understanding and producing human language (speech or text).

Example:

Google Translate converts Bangla ↔ English.

Simple:

Understanding and speaking human language.

How They Work Together — Self-Driving Car

Perception: cameras see a pedestrian



Knowledge: “pedestrians on a crossing have right of way”



Reasoning: plan a path → “slow down and stop”



Machine learning: driving style learned from millions of km of driving data

Easy Memory Trick:

“Kind **M**achines Really Please **N**ations”

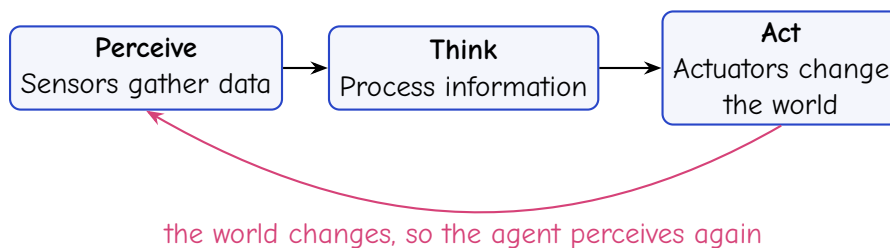
Knowledge, **M**achine learning, **R**easoning, **P**erception, **NLP**

4. Agents and Environments

AI agent = anything that continuously **perceives** its environment, **reasons** about it, and **acts** on it to achieve goals.

Environment = the world around the agent — everything the agent senses and acts upon.

The Perceive → Think → Act Cycle



- **Sensors** = parts that **collect information** (eyes, ears for humans; cameras, microphones for robots).
- **Actuators** = parts that **do things** / cause action (hands, legs for humans; wheels, motors, speakers for robots).

Example:

(Robot vacuum — Roomba)

Perceive: bump sensor touches a wall**Think:** “I can’t go forward, I should turn”**Act:** spins its wheels to turn

It repeats this cycle again and again, all the time it is cleaning.

Think of it like this (real life):

You are a human agent. Your **eyes** see the traffic light turn red (sensor → perceive).
 Your **brain** decides “stop” (think). Your **foot** presses the brake (actuator → act).

PEAS — Describing Any AgentTo design an agent, we describe 4 things. Together they are called **PEAS**:

Letter	Stands for	Question it answers
P	Performance measure	How do we know the agent is doing a good job ?
E	Environment	Where does the agent work?
A	Actuators	What can the agent use to act ?
S	Sensors	What can the agent use to sense ?

Example 1: Self-Driving Taxi (from the lecture)

P — Performance	Safety, speed, comfort, obeying the law
E — Environment	Roads, traffic, pedestrians, weather
A — Actuators	Steering, accelerator, brakes, indicator signals
S — Sensors	Cameras, lidar, radar, GPS

Example 2: Robot Vacuum Cleaner (from the self-check)

P — Performance	Amount of floor cleaned, time taken, battery used, no crashes
E — Environment	Rooms, floors, furniture, walls, stairs, carpets
A — Actuators	Wheels / motors, vacuum suction, brushes
S — Sensors	Bump sensors, dirt sensors, cliff sensors (to detect stairs), cameras

Example 3: Medical Diagnosis System (extra practice)

P — Performance	Correct diagnoses, patient health, low cost
E — Environment	Patients, hospital, doctors, test results
A — Actuators	Screen showing diagnosis, suggested tests, treatment advice
S — Sensors	Keyboard input of symptoms, lab test results, scan images

Common Mistake:

Mixing up **sensors** and **actuators**. Ask: “Does it *bring information in*?” → sensor. “Does it *do something to the world*?” → actuator. A camera is a sensor; a steering wheel is an actuator.

Easy Memory Trick:

PEAS → **P**oints (score), **E**nvironment (place), **A**ctions (hands), **S**enses (eyes)

Types of Environments

Environments are described by 3 main properties:

1. Observable: Fully vs Partially

- **Fully observable** → the agent can see **everything** it needs at every moment.
Example: chess — the whole board is visible.
- **Partially observable** → some information is **hidden or missing**.
Example: poker — you cannot see the other players' cards.

2. Predictability: Deterministic vs Stochastic

- **Deterministic** → the same action **always** gives the same result. No luck involved.
Example: chess — moving a knight always puts it exactly where you move it.
- **Stochastic** → results involve **randomness / chance**. (Stochastic = random.)
Example: poker — which card is drawn next is random. Driving — other drivers are unpredictable.

3. Changes over time: Static vs Dynamic

- **Static** → the world **waits** while the agent thinks.
Example: chess — the board doesn't change until you move.
- **Dynamic** → the world **keeps changing** even while the agent is thinking.
Example: driving — traffic keeps moving, whether you decide or not.

Comparing Three Environments

Property	Chess	Poker	Self-Driving Car
Observable	Fully — whole board visible	Partially — opponents' cards hidden	Partially — blind spots, objects hidden behind others
Predictability	Deterministic — same move, same result	Stochastic — card draws are random	Stochastic — other drivers, weather
Changes over time	Static — waits for your move	Static between turns	Dynamic — traffic keeps moving

Why?

The harder an environment is to **see** and **predict**, the more the agent must use **probability**, **combine many sensors** (sensor fusion), and **keep re-planning**, instead of using one fixed plan.

That's why a **self-driving car is a much harder AI problem than chess**, even though chess "looks" more complex to beginners. Chess is fully observable, deterministic and static — the easiest combination. Driving is the hardest combination of all three.

Easy Memory Trick:

Easy environment = **Fully** observable, **Deterministic**, **Static** (chess)

Hard environment = **Partially** observable, **Stochastic**, **Dynamic** (driving)

5. Fundamental Challenges of AI

These are the **big problems** that make building good AI difficult. There are 5.

1. Data Quality & Bias

AI learns from data. If the data is **unfair or poor**, the AI becomes unfair or poor too.

Biased data → Biased AI ("garbage in, garbage out")

Example:

An automated **hiring tool** was trained mostly on past resumes (which came mostly from men). It learned to give **lower scores** to resumes containing the word "women's" (e.g. "women's chess club captain") — even though that has nothing to do with skill.

Simple:

AI copies the unfairness hidden in its data.

2. Generalization

Generalization = doing well on **new situations** the AI never saw during training.

Example:

Early self-driving car vision models were trained mostly in **sunny** places. The first time they saw **snow-covered** roads, they misjudged the lane markings.

Simple:

AI often fails in situations that are different from its training.

3. Ethics & Safety

Is it **right** and **safe** to use AI this way?

Example:

Facial recognition in public places raises concerns about **privacy** (being watched without permission) and **consent**. It has also been less accurate for people with darker skin tones — which is unfair.

Simple:

Just because AI *can* do something doesn't mean it *should*.

4. Explainability

Explainability = being able to explain **why** the AI made a decision.

Example:

A deep learning model **denies a loan** or **flags a tumour** correctly — but doctors, banks and regulators still need to know **why** before they can trust it. Many models are a “**black box**”: we see the input and output, but not the reasoning inside.

Simple:

A correct answer is not enough; people need to know the reason.

5. Scalability

Scalability = still working well when the number of users or amount of data becomes **huge**.

Example:

A search engine must answer **billions** of searches a day in **milliseconds** — not just work in a small lab demo.

Simple:

Working for 10 people is easy. Working for 1 billion is hard.

Easy Memory Trick:

Challenges = “Don't Give Every Engineer Stress”

Data bias, Generalization, Ethics, Explainability, Scalability

6. Limitations of Today's AI Systems

Challenges are problems in *building* AI. Limitations are weaknesses that today's AI *still has*.

Limitation	Meaning	Example
1. Lack of true understanding	AI processes data but doesn't really understand context or emotion.	A chatbot replies cheerfully to "I lost my job today" without grasping how sad it is.
2. Dependence on data	AI is only as good as the quality and amount of its training data.	A model trained only on English struggles the first time it sees Bangla text.
3. Limited generalization	AI struggles with new, unseen situations much more than humans do.	A vision model trained on day-time photos misreads street signs at night.
4. Energy consumption	Training big models needs huge computing power → lots of electricity.	Training one large language model can use as much electricity as hundreds of homes use in a year.
5. Ethical & bias issues	AI can be biased or confidently wrong.	A generative AI may " hallucinate " — invent a fake book citation that looks completely real.

Hallucination = when AI confidently makes up information that is **false**.

Simple:

Today's AI doesn't truly understand, needs lots of good data, fails in new situations, uses lots of energy, and can be biased or make things up.

7. From Perception to Logic

Humans do something amazing without effort: turn a messy mix of **sights and sounds** into a clear **sentence** and a **decision**. This happens in 4 stages:



- **Perception** = what your senses pick up (raw information).
- **Cognitive integration** = your brain putting all the pieces together. (Cognitive = related to thinking.)
- **Logical sentence** = a clear statement of what is happening.
- **Decision** = what to do about it.

Think of it like this (real life):

You hear a **car horn**, see a **car coming fast**, and feel your friend **pull your arm**. In less than a second your brain says “a car is about to hit me” and you jump back. You didn’t think about each sense separately — your brain combined them instantly.

Why is this hard for AI?

1. **Combining different senses is hard.**
Fusing audio + video into one clear story is still an **open research problem** (not yet solved).
2. **Hidden context is confusing.**
“People running” could mean **danger**, a **marathon**, or kids playing **tag**. Humans use world knowledge to tell the difference; AI often can’t.
3. **Making a natural sentence needs two skills at once.**
Turning the situation into a human-like sentence (not just a label like “fire: 87%”) needs **perception AND language generation** working together.

Simple:

Humans easily turn senses into meaning. For AI, combining senses and understanding context is still very hard.

Self-Check: Environments & Agents

Q1. Is chess fully or partially observable? Deterministic or stochastic?

Ans: Fully observable (the whole board is visible) and **deterministic** (a legal move always has the same result).

Q2. For a robot vacuum cleaner, what are its Sensors and Actuators (PEAS)?

Ans: Sensors: bump sensors, dirt/cliff sensors, cameras. **Actuators:** wheels/motors, vacuum suction, brushes.

8. Why AI Terminology Matters

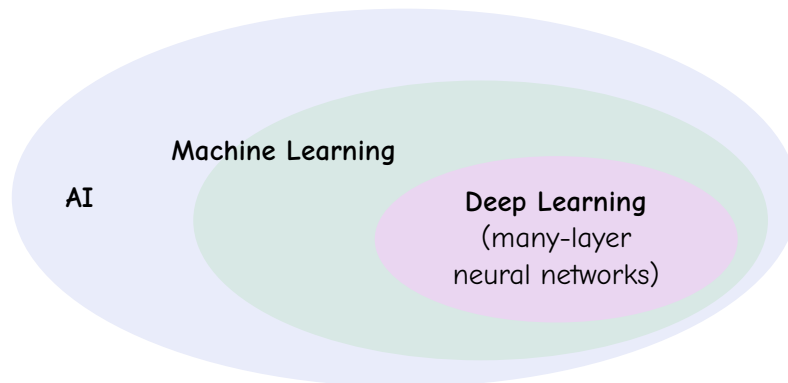
AI is a huge field with many **overlapping** terms (ML, DL, neural networks ... all sound similar). Learning the terms helps you:

- understand the **building blocks** of AI systems,
- grasp AI’s **core principles** and applications,
- **communicate precisely** with other AI practitioners.

9. Core AI Terminology

How the Terms Fit Together

These terms are **nested** — each one is a part of the bigger one:



- **AI** = the whole field of making machines smart (includes search, logic, learning ...).
- **Machine Learning** = the part of AI where machines **learn from data**.
- **Deep Learning** = the part of ML that uses **neural networks with many layers**.

(Computer Vision and NLP are **application areas** — they use ML/DL to work with images and language.)

The 5 Core Terms

1. Neural Network

Neural network = interconnected layers of **nodes** (small calculators), inspired by the neurons in the human brain.

Example: recognising **handwritten digits** (0–9) — the famous **MNIST** dataset.

2. Machine Learning

Machine learning = systems that **learn and improve from data**.

Example: a recommendation engine that gets **better** the more you click.

3. Deep Learning

Deep learning = machine learning using neural networks with **many layers** (“deep” = many layers).

Example: **ResNet** (a famous deep network) classifying images into thousands of categories.

4. Computer Vision

Computer vision = interpreting and making decisions from **visual data** (images, video).

Example: **lane detection** in a self-driving car.

5. Natural Language Processing (NLP)

NLP = understanding and responding to **human language**.

Example: a customer-support chatbot that answers in Bangla or English.

Easy Memory Trick:

AI $\supset$ ML $\supset$ DL $\rightarrow$ like **Country** $\supset$ **City** $\supset$ **Neighbourhood**

Vision = **eyes** of AI NLP = **ears & mouth** of AI

10. From Biological Neuron to Perceptron

The Biological Neuron

A **neuron** is a brain cell. It has 3 main parts:

- **Dendrites** $\rightarrow$ receive signals from other neurons (inputs).
- **Cell body** $\rightarrow$ adds up all the signals.
- **Axon** $\rightarrow$ if the total is strong enough, sends a signal onward (output).

The Perceptron (Artificial Neuron)

Perceptron = the simplest artificial neuron: it takes inputs, multiplies each by a **weight**, adds them up with a **bias**, and decides an output using an **activation function**.

Biological neuron	Perceptron
Dendrites	$\rightarrow$ Inputs (x)
Strength of each connection	$\rightarrow$ Weights (w)
Cell body (adds signals)	$\rightarrow$ Weighted sum (z)
Fires or not	$\rightarrow$ Activation function (f)
Axon (sends signal)	$\rightarrow$ Output (y)

Inputs (x) $\rightarrow$ Weights (w) $\rightarrow$ Weighted Sum (z) $\rightarrow$ Activation (f) $\rightarrow$ Output (y)

What each part means

- **Input (x)** = a number describing something about the data (e.g. brightness of a photo).
- **Weight (w)** = how **important** that input is. Big weight = very important.
- **Bias (b)** = an extra number that shifts the result up or down (like a starting point).
- **Weighted sum (z)** = multiply each input by its weight, add them all, then add the bias:

$$z = (x_1 \times w_1) + (x_2 \times w_2) + b$$

- **Activation function (f)** = the decision rule. A **step** activation says: “if z is above a threshold $\rightarrow$ output 1, otherwise output 0”.

Think of it like this (real life):

Deciding whether to go to a party. Inputs: “my friends are going” (important, high weight), “it’s raining” (less important, low weight). You add up the reasons (weighted sum). If the total is strong enough (above a threshold), you go (output 1); otherwise you stay home (output 0). That’s a perceptron!

Worked Example — One Perceptron Step

Task: decide if a photo is a **cat (0)** or a **dog (1)**.

Label encoding	0 $\rightarrow$ cat, 1 $\rightarrow$ dog
Inputs	$x_1 = 0.8$ (image brightness), $x_2 = 0.3$ (edge count)
Weights	$w_1 = 0.5$, $w_2 = 0.4$
Bias	$b = 0.1$
Activation	Step: output 1 if $z > 0.5$, else 0

Step 1 — Multiply each input by its weight:

$$x_1 \times w_1 = 0.8 \times 0.5 = 0.40$$

$$x_2 \times w_2 = 0.3 \times 0.4 = 0.12$$

Step 2 — Add them and the bias:

$$z = 0.40 + 0.12 + 0.1 = \mathbf{0.62}$$

Step 3 — Apply the activation:

$$0.62 > 0.5 \rightarrow \text{output } y = 1 \rightarrow \text{“dog detected”}.$$

Note: the slide first writes the weighted sum as 0.38, but the correct value is $0.40 + 0.12 + 0.1 = \mathbf{0.62}$ (the slide’s next line also uses 0.62).

Why?

One perceptron is tiny. But **millions** of them connected together, each with its own weights, form the neural networks that power today’s deep learning.

Simple:

A perceptron = multiply inputs by importance, add them up, and say yes/no.

11. Advanced AI Terminology

1. Reinforcement Learning (RL)

Reinforcement learning = an agent learns by **interacting** with an environment, getting **rewards** for good actions and **penalties** for bad ones, and trying to **maximize total reward**.

Think of it like this (real life):

Training a dog: it sits → you give a treat (reward). It jumps on the sofa → you say “No!” (penalty). Over time the dog learns which actions get treats. Nobody told it the rules directly — it learned by trial and error.

Example:

AlphaGo learned to play the board game Go by playing **millions of games against itself**, and went on to beat the world champion.

2. Generative AI

Generative AI = AI that **creates new content** (text, images, music, code) based on patterns learned from training data.

Example:

ChatGPT writes text; **Midjourney** and **DALL·E** create images from a text prompt (“a cat riding a bicycle in Dhaka”).

3. Knowledge Representation

Knowledge representation = structuring information so AI can **reason** over it.

Example:

A medical ontology stores that “fever” is related to “infection”, so the system can reason “patient has fever → check for infection”.

4. Fuzzy Logic

Fuzzy logic = reasoning with **degrees of truth** (anywhere between 0 and 1), instead of strict yes/no (true/false).

Normal logic: a room is either **hot (1)** or **not hot (0)**.

Fuzzy logic: a room can be **“0.6 hot”** — somewhat warm.

Temperature	Normal logic: “Hot?”	Fuzzy logic: “How hot?”
20 °C	No (0)	0.1 (a little)
28 °C	No (0)	0.6 (somewhat warm)
35 °C	Yes (1)	1.0 (fully hot)

Example:

A smart air conditioner decides the room is **“somewhat warm”** and cools **gradually**, instead of switching fully ON or fully OFF.

Simple:

Fuzzy logic = “a little”, “somewhat”, “very” — not just yes or no.

5. Swarm Intelligence

Swarm intelligence = smart group behaviour that comes from many simple agents working together **without a central leader** (decentralized).

Think of it like this (real life):

A single ant is not smart. But a whole ant colony finds the **shortest path** to food: ants leave a scent trail, shorter paths get more scent faster, and more ants follow them. No ant is “in charge”.

Example:

- **Ant-colony optimization** is used to find good routes for delivery trucks.
- **Drone swarms** fly together in formation (e.g. drone light shows).

Easy Memory Trick:

RL → **Rewards** GenAI → **Creates** KR → **Organised facts** Fuzzy → **Degrees**
 Swarm → **Team of simple agents**

12. Evaluation Metrics — How Good is the AI?

Evaluation metric = a number that tells us **how well** an AI model performs.

The Confusion Matrix

When an AI answers “yes/no” questions (e.g. “Is this email spam?”), every answer falls into one of 4 boxes:

	AI says: Spam	AI says: Not Spam
Really Spam	TP — True Positive	FN — False Negative
Really Not Spam	FP — False Positive	TN — True Negative

- **True Positive (TP)** → it was spam, AI said spam. **Correct**.
- **True Negative (TN)** → it was a normal email, AI said not spam. **Correct**.
- **False Positive (FP)** → it was a normal email, AI wrongly said spam. **Wrong** (false alarm).
- **False Negative (FN)** → it was spam, AI wrongly said not spam. **Wrong** (missed it).

Easy Memory Trick:

Second word = what the AI **said** (Positive = “yes, spam”, Negative = “no”).
 First word = was the AI **right**? (True = right, False = wrong).

Worked Example — Spam Filter

A spam filter checks **100 emails**:

- Correctly flags **30** real spam emails → TP = 30
- Misses **10** spam emails (calls them safe) → FN = 10
- Wrongly flags **5** real emails as spam → FP = 5
- The rest are correctly called safe: $100 - 30 - 10 - 5 = 55 \rightarrow$ TN = 55

	Predicted: Spam	Predicted: Not Spam
Actual: Spam (40)	30 (TP)	10 (FN)
Actual: Not Spam (60)	5 (FP)	55 (TN)

1. Accuracy

Accuracy = out of **all** predictions, how many were correct?

$$\text{Accuracy} = (\text{TP} + \text{TN}) / \text{Total} = (30 + 55) / 100 = 85\%$$

2. Precision

Precision = when the AI says “spam”, how often is it **really** spam?

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) = 30 / (30 + 5) = 30 / 35 = 85.7\%$$

High precision = **few false alarms**.

3. Recall

Recall = out of all the **real** spam, how much did the AI **catch**?

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) = 30 / (30 + 10) = 30 / 40 = 75.0\%$$

High recall = **few misses**.

4. F1-Score

F1-score = one number that **balances** precision and recall (their harmonic mean).

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$\text{F1} = 2 \times (0.857 \times 0.75) / (0.857 + 0.75) = 1.286 / 1.607 \approx 80.0\%$$

F1 is high only when **both** precision and recall are high.

Think of it like this (real life):

A **fishing net**. **Recall** = how many of the fish in the lake did you catch? **Precision** = of everything in your net, how much is actually fish (and not old shoes)? A huge net catches all the fish (high recall) but also lots of rubbish (low precision).

Why Not Just Use Accuracy?

Accuracy here is 85% — sounds great! But the filter still **misses 1 in 4 real spam emails** (recall = 75%). Accuracy hides this.

On **imbalanced data** (one class is much rarer than the other), accuracy can be **misleading**. Example: if only 2% of patients have a disease, a useless model that says “healthy” to **everyone** gets **98% accuracy** — but catches **zero** sick patients (recall = 0%). Precision, Recall and F1 show the **real story**.

When to care more about which?

- **Recall matters more** when **missing** something is dangerous.
Example: cancer detection — better a false alarm than missing a real cancer.
- **Precision matters more** when a **false alarm** is costly.
Example: spam filter — sending an important job-offer email to spam is very bad.

Easy Memory Trick:

Precision → “When I say **Positive**, am I right?”

Recall → “Did I **Remember** (catch) them all?”

Other Key Terms

Overfitting

Overfitting = a model that **memorizes** the training data but performs **poorly on new data**.

Think of it like this (real life):

A student memorizes the exact answers to last year’s exam questions instead of understanding the concepts. If the same questions come, they get full marks. If the questions change even a little, they fail. That student has **overfitted**.

Training data: 99% correct but New data: 60% correct → Overfitting

Mean Squared Error (MSE)

MSE = the **average of the squared errors** between predictions and real values. Used when predicting **numbers** (called **regression**), e.g. house prices.

Error = Real value – Predicted value → square it → take the average

Example:

(small, for understanding) Predicting house prices (in lakh Tk):

Real	Predicted	Error	Error ²
50	48	2	4
70	74	-4	16
60	60	0	0

$$\text{MSE} = (4 + 16 + 0)/3 \approx \mathbf{6.67}.$$

Why?

(square the errors?) So that negative and positive errors don't cancel out, and so that **big mistakes are punished more**.

Simple:

MSE = on average, how far off are the number predictions? Smaller = better.

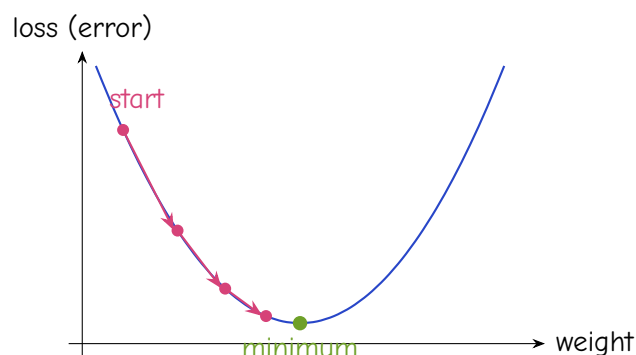
13. Gradient Descent

Gradient descent = an optimization method that repeatedly **nudges** a model's weights **downhill**, toward the **lowest error** (called **loss**).

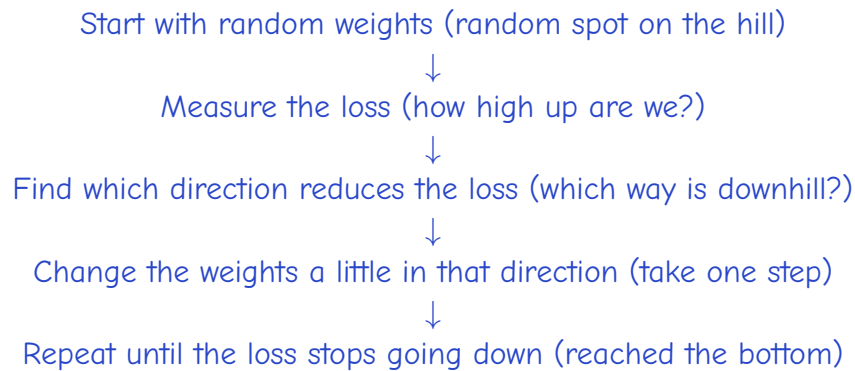
Loss = how wrong the model is. Training = making the loss as small as possible.

Think of it like this (real life):

You are standing on a **hill in thick fog** and want to reach the **lowest point** of the valley. You can't see far, but you can feel which way the ground slopes under your feet. So you take a step **downhill**, feel again, take another step downhill ... until the ground is flat. That's gradient descent.



How It Works



Each step = **one training update**.

Learning Rate

Learning rate = the **size of each step** in gradient descent.

Learning rate	What happens
Too small	Tiny baby steps → reaches the bottom, but very slowly (takes forever to train).
Just right	Good steps → reaches the bottom quickly and safely .
Too large	Huge leaps → jumps over the bottom again and again, bouncing from side to side, and may never settle (fails to converge).

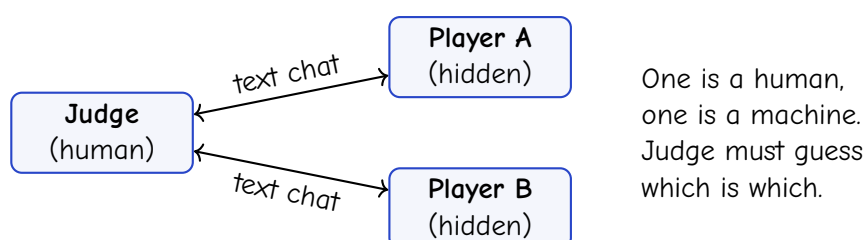
Converge = settle down at the lowest point.

Simple:

Gradient descent = keep taking small steps downhill to reduce the error. Learning rate = step size.

14. The Turing Test

Turing Test = a test of machine intelligence proposed by **Alan Turing** (1950). A human judge chats by **text** with both a human and a machine, without knowing which is which. If the judge **cannot reliably tell them apart**, the machine is said to have **passed**.



Example:

Judge: “What’s it like to taste coffee for the first time?”

Player A (machine): “Bitter, warm, a little bracing — it wakes you up before you even finish the sip.”

Player B (human): “Honestly kind of gross at first, then you crave it every morning.”

Both answers sound human! It’s hard to tell which is the machine.

Is the Turing Test a good measure of intelligence?

Modern large language models (like ChatGPT) easily produce answers like Player A. So today the Turing Test is **debated**:

- A machine can **sound** human without **understanding** anything (it has never tasted coffee!).
- It tests **imitation of conversation**, not true intelligence, feelings, or understanding.

Simple:

Turing Test = can a machine chat so well that a human can’t tell it’s a machine? Passing it doesn’t prove real understanding.

Self-Check: Metrics & Concepts

Q1. A disease-detection model has 95% accuracy, but the disease is rare (2% of patients). Is accuracy the best metric?

Ans: No. With imbalanced classes, a model can get high accuracy just by saying “healthy” to everyone. Precision, Recall and F1-score show how well it actually **catches** the disease.

Q2. In gradient descent, what happens if the learning rate is far too large?

Ans: The model **overshoots** the minimum again and again and **fails to converge** — like taking huge leaps down a hill and bouncing past the bottom instead of settling into it.

15. Key Takeaways

Remember

- **Intelligence** = acquire, process, apply knowledge + adapt.
- **AI** simulates human intelligence but doesn't truly "understand" like people.
- Humans: few examples, context, emotion. AI: speed, scale, consistency.
- **5 components**: Knowledge representation, Machine learning, Reasoning, Perception, NLP.
- **Agent** = Perceive → Think → Act. Sensors bring info in; actuators act.
- **PEAS** = Performance, Environment, Actuators, Sensors.
- Environments: Fully/Partially observable, Deterministic/Stochastic, Static/Dynamic.
- Driving is harder than chess (partial, stochastic, dynamic).
- **Challenges**: data bias, generalization, ethics, explainability, scalability.
- **Limitations**: no true understanding, data dependence, poor generalization, energy use, bias/hallucination.
- **Perception** → **Logic**: raw senses → integration → sentence → action. Hard for AI.
- $AI \supset ML \supset \text{Deep Learning}$ (many-layer neural networks).
- **Perceptron**: $z = \sum x_i w_i + b$, then activation → output.
- RL = rewards; GenAI = creates; Fuzzy = degrees of truth; Swarm = decentralized group.
- Accuracy = $(TP+TN)/\text{Total}$; Precision = $TP/(TP+FP)$; Recall = $TP/(TP+FN)$; F1 = balance.
- **Overfitting** = memorizing, not learning. **MSE** = average squared error.
- **Gradient descent** = step downhill on the loss; learning rate = step size.
- **Turing Test** = can a judge tell human from machine by text chat?

16. Exam Questions

Q1. Define intelligence and AI. Give 3 differences between human and artificial intelligence.

Ans: Intelligence is the ability to acquire, process and apply knowledge to solve problems and adapt to new situations. AI is the simulation of human intelligence in machines to perform tasks like learning, reasoning, problem-solving and decision-making. Differences: (1) humans learn from few examples, AI needs large datasets; (2) humans understand sarcasm and emotion, AI struggles without explicit signals; (3) AI is faster and never tired, humans are limited by biology.

Q2. Write the PEAS description of a self-driving taxi.

Ans: Performance: safety, speed, comfort, legality. Environment: roads, traffic, pedestrians, weather. Actuators: steering, accelerator, brakes, signals. Sensors: cameras, lidar, radar, GPS.

Q3. Why is self-driving harder than chess for AI?

Ans: Chess is fully observable, deterministic and static — the agent sees everything, moves always have the same result, and the board waits. Driving is partially observable (blind spots),

stochastic (unpredictable drivers, weather) and dynamic (traffic keeps moving). So the car must use probability, sensor fusion and constant re-planning.

Q4. Explain data bias with an example.

Ans: AI learns from data, so unfair patterns in the data become unfair decisions. A hiring tool trained mostly on men's resumes learned to downrank resumes containing "women's" (e.g. "women's chess club").

Q5. Compute the perceptron output: $x_1 = 0.8$, $x_2 = 0.3$, $w_1 = 0.5$, $w_2 = 0.4$, $b = 0.1$, **threshold 0.5.**

Ans: $z = 0.8(0.5) + 0.3(0.4) + 0.1 = 0.40 + 0.12 + 0.1 = 0.62$. Since $0.62 > 0.5$, $y = 1 \rightarrow \text{dog}$.

Q6. A spam filter has TP = 30, FN = 10, FP = 5, TN = 55. Find accuracy, precision, recall and F1.

Ans: Accuracy = $85/100 = 85\%$. Precision = $30/35 = 85.7\%$. Recall = $30/40 = 75\%$. $F1 = 2(0.857 \times 0.75)/(0.857 + 0.75) \approx 80\%$.

Q7. What is overfitting?

Ans: When a model memorizes the training data instead of learning general patterns, so it performs well on training data but poorly on new data — like memorizing exam answers instead of understanding the concept.

Q8. Explain gradient descent and the role of learning rate.

Ans: Gradient descent repeatedly adjusts the model's weights in the direction that reduces the loss, like walking downhill to the lowest point. The learning rate is the step size: too small is very slow, too large overshoots and may never converge.

Q9. What is the Turing Test? Why is it debated today?

Ans: A judge chats by text with a hidden human and a hidden machine. If the judge can't reliably tell which is the machine, the machine passes. It is debated because modern language models can imitate human conversation very well without truly understanding anything.

Practice Questions

1. Compare human and artificial intelligence using 5 traits.
2. Explain the sarcasm example. Why does a basic text classifier fail?
3. List and explain the 5 components of AI with one example each.
4. What is an agent? Explain the perceive–think–act cycle with a robot vacuum.
5. Write PEAS for: (a) a robot vacuum, (b) a medical diagnosis system, (c) a chess-playing program.
6. Explain fully/partially observable, deterministic/stochastic, static/dynamic with examples.
7. Explain the 5 fundamental challenges of AI with examples.
8. What are the limitations of today's AI? What is hallucination?
9. Explain the 4 stages from perception to logic. Why is this hard for AI?
10. Draw the relationship between AI, ML and Deep Learning.
11. Map the parts of a biological neuron to a perceptron.
12. Define reinforcement learning, generative AI, fuzzy logic and swarm intelligence with examples.

13. Draw a confusion matrix and define TP, TN, FP, FN.
14. Why can accuracy be misleading on imbalanced data?
15. When is recall more important than precision? Give an example.
16. Calculate MSE for real values (10, 20) and predictions (12, 17). (Ans: $(4 + 9)/2 = 6.5$)
17. What happens if the learning rate is too small or too large?
18. Explain the Turing Test with an example.